

Jonah Sussman

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Work Experience

Red Hat - Raleigh, NC (Remote) | Senior Software Engineer, Migration | 05/2022 - Present

- **Founded and led development of [Kai](#) and [migeval](#)**, an agentic & RAG-based **LLM-driven migration tool** that uses [analyzer-lsp](#) & [tree-sitter](#) to support AI-assisted code migration, in **TypeScript** and **Python**
- Designed and implemented **MCP-oriented** concepts for storing, querying, and reusing migration solutions, agent trajectories, and internal developer knowledge across AI coding workflows.
- **Presented Kai at KubeCon EU 2025** and engaged directly with the open-source community to drive adoption and collect feedback
- **Engaged** in Red Hat's internal contribution initiatives for **PyTorch**, **vLLM**, and **Triton**
- **Expanded [analyzer-lsp](#)**, a **language-server-based static analysis engine**, by contributing core improvements and enhancing YAML-based rule definitions across **Java**, **Python**, **C#**, and **Go**
- **Built [sequoia-diff](#)**, an **AST diffing engine** in **Python** that computes tree edits and code similarity, enabling intelligent fix suggestion retrieval across large codebases
- **Partnered cross-functionally** with PMs and customers to shape AI tooling around real-world developer workflows and migration pain points

Digital Forensics & Security Lab - Tuscaloosa, AL | Research Assistant | 09/2020 - 04/2022

- **Published a paper** on [LIDATS](#), an **IoT** system using **Go** and **Rust** to monitor traffic patterns for ALDOT
- Researched detecting malicious Android software with **PyTorch**, achieving an 85% detection rate

FiServ - Alpharetta, GA | Summer Intern | 06/2021 - 08/2021

- Built a **.NET** and **SQLite**-based tool notifying SREs of CVEs, reducing triage time from hours to minutes

Projects

[gpunix](#) - Linux kernel running entirely on GPU. Built from scratch to explore CUDA GPU control flow (C++17)

[aeditor](#) - Custom AI-powered video editor that uses forced alignment to isolate and reorder spoken words in videos. Built custom audio and video rendering engine to overcome limitations in industry tools (C++11)

[software-renderer](#) - 3D software renderer with custom SIMD linear algebra library (C++11)

Skills

AI - LLMs, agentic workflows, MCP, coding assistants, prompt engineering, LLM evaluation, retrieval-augmented generation, semantic search, context construction, tool orchestration

Languages - Python, TypeScript, C, C++, Go, Rust, Java

Systems & Infra - Kubernetes, Docker, vLLM, PyTorch, distributed systems, API & data pipeline design

Education

The University of Texas at Austin - Austin, TX (Remote)

MS, Artificial Intelligence - Spring 2027 (Expected)

Relevant coursework: Machine Learning, Natural Language Processing, Advances in Deep Learning

The University of Alabama - Tuscaloosa, AL

MS, Computer Science - Spring 2023 (GPA: 3.9/4.0)

BS, Computer Science & Applied Mathematics - Fall 2022 (GPA: 4.0/4.0)